General Binomial Expansion

Exam code: 9MA0

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General binomial expansion

What is the general binomial expansion?

- The binomial expansion applies for positive integers, $n \in \mathbb{N}$
- The **general binomial expansion** applies to other types of powers too

THE GENERAL BINOMIAL THEOREM

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

FOR $n \in \mathbb{R}$ AND $|x|<1$

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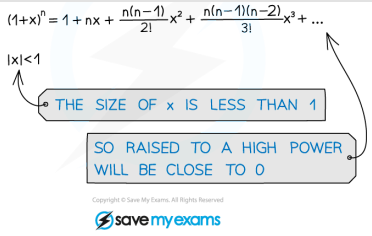
- The **general binomial expansion** applies for all real numbers, $n \in \mathbb{R}$
- Usually fractional and/or negative values of **n** are used
- It is derived from $(a + b)^n$, with $a = 1$ and $b = x$
- $a = 1$ is the main reason the expansion can be reduced so much

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



- Unless $n \in \mathbb{N}$, the expansion is infinitely long
- It is only valid for $|x| < 1$
  - This is another way of writing $-1 < x < 1$
  - This is often called the **validity statement**
  - The restriction $|x| < 1$ means the **series** will **converge**
  - Higher powers of x can be ignored (as $r \to \infty$, $x^r \to 0$)
  - Only the first few terms of an expansion are needed

How do I apply the general binomial expansion?

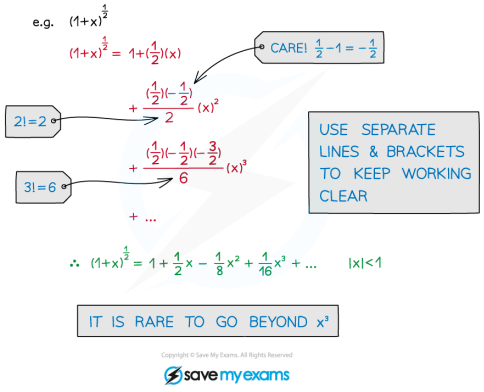
STEP 1 Write the expression in the form $(1 + x)^n$

STEP 2 Expand and simplify

Use a line for each term to make things easier to read and follow

Use brackets - fractions and negatives get ugly!

STEP 3 If required, check and state the validity statement



How do I use the general binomial expansion to expand $(1+bx)^n$?

STEP 1 Write the expression in the form $(1 + bx)^n$

STEP 2 Replace "x" by "bx" in the expansion

Check carefully to see if b is negative

STEP 3 Expand and simplify

Use a line for each term to make things easier to read and follow

Use brackets

Replace **x** with **bx** so now **|bx| < 1**

e.g. FIND THE FIRST THREE TERMS IN THE EXPANSION OF $\frac{1}{\sqrt{1-x}}$

STEP 1: $\frac{1}{\sqrt{1-x}} = (1-x)^{-\frac{1}{2}}$

WRITE IN THE FORM $(1+bx)^n$

STEP 2: $= 1 + (-\frac{1}{2})(-x) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2}(-x)^2 + \ldots$

REPLACE "x" WITH "−x"

STEP 3:

$\therefore \frac{1}{\sqrt{1-x}} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \ldots$

EXPAND AND SIMPLIFY

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e.g. FOR WHICH VALUES OF x IS THE EXPANSION OF $\frac{1}{\sqrt{1-x}}$ VALID?

STEP 4: $\frac{1}{\sqrt{1-x}} = (1-x)^{-\frac{1}{2}}$

VALID WHEN $|-x|<1$

REPLACE "x" WITH "−x"

$\therefore$ VALID WHEN $|x|<1$

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